

Eesha Tolety

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CAREER OBJECTIVE

I graduated with a Master's in aerospace engineering from CU Boulder, specializing in structures, fluids, and materials. I led the structural design and FEA validation for FLAIR, a fixed-wing UAV payload project at CU Boulder, verified it through cold-environment and mechanical integration testing, and built a CFD solver in a smaller course project using FEM discretization and GMSH meshing. I like taking design ideas from concept to CAD and develop models that are engineered, tested, and aligned with real-world constraints.

TECHNICAL SKILLS

- **Software:** SolidWorks, ABAQUS, ANSYS(Structural, Fluent, CFX, Workbench, APDL), OpenFOAM, Paraview, Solid Edge, NX, Creo, Onshape, AutoCAD, Rhino, Prusa Slicer, 3-D Printing Slicers.
- **Programming Languages:** MATLAB, MATLAB Simulink, Python, C++,

EDUCATION

- **Master of Science in Aerospace Engineering** **August 2024 - May 2026**
University of Colorado Boulder
Fluids, Structures and Materials
GPA: 3.67/4.00
- **Bachelor of Science in Aerospace Engineering** **July 2019 - June 2023**
Amity University
GPA: 8.74/10.00 Grade: A

PROJECTS

- **FLAIR** **August 2025 - May 2026**
Graduate Project, University of Colorado Boulder
Boulder, CO
This project involves incorporating a new type of payload into a large VTOL drone. As one of the structural leads on the project, I have been using SolidWorks and Solid Edge for the design work. Additionally, I utilized SolidWorks and Abaqus for design analysis. During the production phase, I assisted with fabrication and testing of the components before the final flight.
- **Microstructural and mechanical analysis of multi-walled Carbon Nanotube(CNTs) and Magnesium(Mg) composite.** **January 2023 - June 2023**
Major Project, Amity University
Noida, India
A CNT-reinforced magnesium composite was developed using a modified friction stir processing technique to improve CNT dispersion. XRF confirmed the presence of carbon, magnesium, and oxygen, while SEM showed a consistent distribution of CNTs across the sample. Mechanical properties, such as macro-hardness, were also investigated.
- **Overall Enhancement of Physio-mechanical characteristics of Composite materials produced with Carbon Nanotube (CNTs)** **August 2022 - January 2023**
Minor Project, Amity University
Noida, India
In this study, pure magnesium was reinforced with varying weight percentages (0.3%–1.8%) of multiwalled CNTs, prepared from organic waste like sugarcane, using stir casting. Mechanical properties were significantly improved, with the optimal CNT content found between 1.2-1.5% by weight for overall property enhancement.

EXPERIENCE

- **University of Colorado Boulder** **January 2026 - May 2026**
Teaching Facilitator: Foundations of Propulsion
Boulder, CO
Assisting students in resolving questions related to numerical problems and conceptual understanding. Evaluating homework assignments and examinations. Organizing study halls to facilitate a deeper comprehension of the subject matter.
- **The Digital Media Center** **January 2026 - May 2026**
Student Leader
Boulder, CO
Overseeing student workers, managing communication between staff and management, and improving 3-D printing workflow and services.
- **Aeromodelling Club** **January 2025 - January 2026**
Assistant technician
Supporting technical operations, equipment troubleshooting, and hands-on fabrication assistance for the 3-D printing lab.
- **Aeromodelling Club** **August 2022 - June 2023**
Vice-President, Amity University
Noida, India
Preparing for competitions and managing inventory for model plane construction. Conducting workshops to share knowledge on designing and flying drones and model planes.

- **Hindustan Aeronautics Limited**

May 2022 - June 2022

Internship, Study of In-flight Refueling System

Nasik, India

In this study, I gained detailed information about the in-flight refueling system, including the necessary components for the process and how it is executed. Through my internship in a comprehensive shop, I was able to observe the entire operation in detail and gain real-life experience, as well as identify the problems that can arise during this process.

ACHIEVEMENTS

- Tolety, E., Gupta, A., Singh, N., Saluja, R.K., Gahlot, N.K. (2024). Microstructural Analysis of Mg/CNT Surface Composite Using Friction Stir Processing. Springer, Singapore.

https://doi.org/10.1007/978-981-97-1306-6_15

INVOLVEMENT

- **Jax Foundation**

August 2022 - December 2022

Volunteer teacher, providing basic English, Science, and Mathematics instruction to underprivileged children.

- **Aeromodelling Club**

August 2019 - June 2023

I began as a club member and became Vice-President in my final undergraduate year.